Header Section

Qarabağ University

Faculty of MATH and IT

Laboratory Assignment № 1

Topic: Conditional Statements and Decision-Making in Python (`if`, `elif`, `else`)

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Task Descriptions

Task 1: Body Mass Index (BMI) Calculator

Objective: Calculate the user's Body Mass Index (BMI) based on entered weight in kilograms and height in meters, and determine the corresponding health category.

Working Principle: The program uses the standard formula BMI = weight/(height*height). Based on the result, it uses `if-elif-else` conditional statements to classify the user as Underweight, Normal Weight, Overweight, or Obese.

Task 2: Chinese Zodiac Calculator

Objective: Determine the Chinese Zodiac animal sign based on the user's birth year.

Working Principle: The Chinese zodiac follows a 12-year cycle. The program calculates the remainder of the year divided by 12 (`year % 12`). Depending on the remainder (0 to 11), it displays the corresponding animal such as Monkey, Rooster, Dog, Pig, Rat, Ox, Tiger, Rabbit, Dragon, Snake, Horse, or Sheep.

Task 3: Birthday Guessing Game

Objective: Guess the user's birthday date of the month logically using 5 predefined number sets without asking the date directly.

Working Principle: The program displays 5 separate number sets and asks whether the birthday is included in each set (`1` for Yes, `0` for No). It sums up the first numbers of the sets where the user answered "Yes" to accurately calculate and display the birth date.

```python
weight = float(input("Enter weight: "))
height = float(input("Enter height: "))
bmi = weight / (height * height)
print(bmi)
if bmi < 18.5:
    print("Category: Underweight")
elif  bmi > 18.5 and bmi < 24.9:
    print("Category: Normal Weight")
elif bmi > 25 and bmi < 29.9:
    print("Category: Overweight")
else:
    print("Category: Obese")
```

```python
set1 = [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
set2 = [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
set3 = [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
set4 = [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
set5 = [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
birthday = 0
a = int(input("Is your birthday included in this set1: 1 for Yes, 0 for No"))
if a == 1:
    birthday = birthday + 1
a = int(input("Is your birthday included in this set2: 1 for Yes, 0 for No"))
if a == 1:
    birthday = birthday + 2
a = int(input("Is your birthday included in this set3: 1 for Yes, 0 for No"))
if a == 1:
    birthday = birthday + 4
a = int(input("Is your birthday included in this set4: 1 for Yes, 0 for No"))
if a == 1:
    birthday = birthday + 8
a = int(input("Is your birthday included in this set5: 1 for Yes, 0 for No"))
```

```
"C:\Users\Asus Vivobook\PycharmProjects\W
Enter your birth year: 2009
Ox

Process finished with exit code 0
```

```python
year = int(input("Enter your birth year: "))
remainder = year % 12
if remainder == 0:
    print("Monkey")
elif remainder == 1:
    print("Rooster")
elif remainder == 2:
    print("Dog")
elif remainder == 3:
    print("Pig")
elif remainder == 4:
    print("Rat")
elif remainder == 5:
    print("Ox")
elif remainder == 6:
    print("Tiger")
elif remainder == 7:
    print("Rabbit")
elif remainder == 8:
```

```
Is your birthday included in this set2: 1 for Yes, 0 for No1
Is your birthday included in this set3: 1 for Yes, 0 for No1
Is your birthday included in this set4: 1 for Yes, 0 for No0
Is your birthday included in this set5: 1 for Yes, 0 for No0
Your birthday is: 6

Process finished with exit code 0
```

Run labbmi

```
"C:\Users\Asus Vivobook\PycharmProjects\Welcom
Enter weight: 75
Enter height: 1.78
23.671253629592222
Category: Normal Weight

Process finished with exit code 0
```